

ExamShield - Business Solution & Future State

Proposed operating model for examination integrity, centre risk and incident management

1. Problem statement

Examination authorities already operate multiple security controls, but a portfolio analysis suggests a practical visibility problem: control information, incidents, candidate impact and recovery actions can sit across different systems or agencies. The proposed solution is therefore not 'more CCTV' by itself. It is **better orchestration, exception visibility and auditability**.

2. Stakeholder map

Stakeholder	Need from analytics	Action
Examination Controller	System-wide integrity and exception view	Escalate / allocate resources
Regional Coordinator	Centre clusters and recurring issues	Regional corrective action
Centre Superintendent	Live centre exceptions	Fix / document / escalate
IT Operations	Network, CCTV, biometric failures	Restore service / create incident
Security & Surveillance	Custody and monitoring controls	Investigate control breaches
Compliance / Audit	Evidence, logs and closure	Audit and verify CAPA
Candidate Support	Complaints and candidate impact	Communicate and resolve

3. Current state

Planning -> Centre allocation -> Secure custody -> Transport -> Centre receipt -> Candidate verification -> Exam conduct -> Incident reporting -> Escalation -> Investigation -> Recovery/re-examination -> Audit closure

Observed from public materials: NTA describes multiple security layers; NTA also maintains re-examination and biometric-exception document trails; SSC reporting shows a candidate-impact data gap around centre changes and technical/logistical failures.

4. Future state

Single control tower -> live/near-live exception feeds -> centre risk score -> threshold alert -> incident owner -> SLA timer -> corrective action -> evidence attachment -> audit closure -> candidate impact view

5. Proposed requirements

ID	Requirement	Priority
FR-01	Dashboard shall show centre risk, incident count, candidate impact and unresolved exceptions.	High
FR-02	Dashboard shall allow drill-down by exam, date, region and centre.	High
FR-03	Every exception shall have an owner and response SLA.	High
FR-04	Control evidence shall be traceable to a source/system record.	High
FR-05	Biometric/CCTV/network exceptions shall be logged with resolution status.	Medium
FR-06	Candidate impact shall be captured as a separate measurable field.	High

ID	Requirement	Priority
FR-07	Risk thresholds shall be configurable and documented.	Medium
FR-08	Closed incidents shall retain an audit trail and corrective-action record.	High

6. KPI catalogue

Centre Risk Score | Critical Exception Count | CCTV Availability | Biometric Exception Rate | Network Downtime | Chain-of-Custody Completeness | Incident Response SLA | Candidate Impact | Complaint Volume | Re-examination Burden | Corrective Action Closure Rate

7. Business impact

The value is better prioritization rather than a simplistic 'high-risk centre' label. A control tower can help operations teams see where an exception is unresolved, assign ownership, measure response time, and preserve an auditable evidence trail.

8. Implementation roadmap

Phase	Deliverable	Output
1 - Discover	Stakeholder interviews + process map	BRD + AS-IS
2 - Define	Data dictionary + KPI definitions + risk logic	Data model
3 - Build	SQL + Power BI + exception workflow	BI report
4 - Validate	UAT + reconciliation against source records	UAT sign-off
5 - Deploy	Refresh schedule + governance + ownership	Control tower